



First Semester 2026-27

COURSE HANDOUT (Part-II)

In addition to part-I (General Handout for all courses) printed on page 1 of the timetable book, this portion gives further specific details regarding the course.

Course Number : **BIO F213**
Course Title : **CELL BIOLOGY**
Instructor-in-Charge : **Syamantak Majumder**
Co-Instructor : **Shashi Prakash Singh**

1. Course Outline:

Fundamental processes of life at cellular and sub-cellular levels; cell environments; membrane transport; cell movements; division and control mechanisms.

2. Scope and Objective:

A fundamental branch in biological sciences, cell biology exposes the learner to the structural and functional organization and working of the cell, the basic unit of life. The field is dynamic and constantly evolving. This course will give an introduction to this branch of biology, by providing experimental data wherever possible. Some of the major topics covered include the working of the cell at microscopic and molecular levels, their physiological properties, structure, intracellular organelles, interactions with microenvironments, division, regulatory mechanisms and related experimental procedures. The topics learned in this course will also be foundational for advanced courses in cell and molecular biology, as well as genetics and developmental biology.

3. Text Book (TB):

Sheeler, P. and Bianchi, DE. (2009). *Cell and Molecular Biology (3/E)*. New Delhi: Wiley.

4. Reference Books (RB):

RB1: Alberts, B., et al. (2014). *Essential Cell Biology (4/E)*. New York: Garland Science.

RB2: Cooper, G.M. and Hausman, R.E. (2015). *The Cell: A Molecular Approach (7/E)*. United States: Sinauer Associates Inc.

RB3: Lodish, H., Berk, A., Kaiser, C.A., Kreiger, M. and Bretscher, A. (2016) *Molecular Cell Biology (8/E)*. United States: W H Freeman & Co.

RB4: Hardin, J., Bertoni, G. and Kleinsmith, L.J. (2016). *Becker's World of the Cell (8/E)*. Essex: Pearson Education Limited.



5. Tentative Lecture Plan:

Lecture No	Topic	Details of coverage	References
L1	Orientation and Introduction	Orientation to the pedagogy, course administration and the class; overview of cell organelles; the composite animal, plant and bacterial cells; mycoplasma and viruses	Ch.1 (TB) Ch. 1 (RB1)
L2-L5	Microscopy	Light, fluorescent and confocal and electron microscopy; preparation of sample for microscopy and specialized applications.	Ch.1 (TB) Ch. 1 (RB1)
L6-L11	Cell Membrane and cell-cell junctions	Structure and chemical organizations of plasma membrane; lipids, carbohydrates and proteins in the membrane; membrane fluidity and lipid asymmetry; cell-cell junctions and other specialized structures.	Ch. 15 (TB) Ch. 11 (RB1)
L12-L16	Membrane functions	Passive movement through cell membrane, facilitated diffusion, active transport, bulk transport, endo- and exocytosis; membrane channels, channel inhibitors and their therapeutic importance.	Ch. 15 (TB) Ch. 12 (RB1)
L17-L21	Endomembrane system	Membrane-enclosed organelles, protein trafficking and transportation, post- translational modifications	Ch. 15, 16,18 (TB) Ch. 15 (RB1)
L22-L25	Cytoskeleton	Cytoplasmic filaments, microtubules, spindle fibers and centriole structures and functions	Ch. 23 (TB) Ch. 17 (RB1)



L26-L30	Cell cycle	Overview of the cell cycle, regulation of cell cycle; cell cycle and cell division	(Ch. 18 (RB1))
L31-L35	Apoptosis	Mechanism of programmed cell death/apoptosis	Ch. 18 (RB1)
L36-L40	Cell communication	General principles; signaling molecules, receptors, secondary messengers, signal transduction, receptor-mediated signaling	Ch. 16 (RB1)

6. Evaluation Scheme:

Component	Weightage	Duration	Date and Time	Remarks
Mid Semester Exam	50 (25%)	90 min	08/10 AN1	Closed Book
Course Quizzes (2) (Announced)	40 (20%)	-	TBA	Closed Book
Tutorial Lab Activity and Assessment (Announced)	20 (10%)	-	During Tutorial Hours	Open Book
In-class Assignment	20 (10%)	-	To be held periodically, mainly unannounced	Closed Book/Open Book
Comprehensive Examination	70 (35%)	180 min	12/12 FN	Closed Book/Open Book

7. **Academic Conduct Policy:** It is expected that all students follow the highest standard of academic practice when participating in any evaluation component. Having a zero-tolerance for academic dishonesty, any case of misconduct, however minor, will be dealt with strictly with necessary actions and the case may be reported to the AUGSD, if necessary.



Criterion for NC and Grading Policy: Award of grades would be guided by the histogram of marks and course average. If a student was found absent himself/herself in any component (listed in the Evaluation Scheme), his/her performance in the course may be reported as 'NC' (Not Cleared). For a student on the borderline of two grades, the decision on which grade to award will depend on the Instructor's recommendation (considering student's regularity/sincerity in classes and completing assigned tasks).

Chamber Consultation: Will be discussed in the class

Announcements and Notices: All announcements regarding the course will be made either through ERP, Nalanda or through emails.

Make-up policy: If a student misses the Mid-Semester or Comprehensive Examination due to a genuine reason (serious medical causes leading to hospitalization (certified by doctor from BITS medical center), personal/family emergencies) there exists a provision to apply for make-up. Prior permission must be taken from the Instructor-in-Charge, before applying for make-up. Make-up decisions will be made on a case-by-case basis and only genuine cases as determined by the Departmental team will be considered. No Make-Up will be granted for quiz, assignments and project assignments.

Grading Policy: If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student will not be awarded a valid grade and the case will be reported as "NC".

Note: It shall be the responsibility of the individual student to be regular in attending lectures and tutorials as per the schedule announced.

Instructor-in-Charge
BIO F213